Host & Network Penetration Testing: The Metasploit Framework CTF 1 Report

Objective: This report documents the steps performed during the CTF lab "Host & Network Penetration Testing: The Metasploit Framework CTF 1". The goal was to gain access to a Windows target machine, exploit an MSSQL service, obtain a Meterpreter session, and retrieve multiple flags hidden within the system directories.

Step 1: Initial Access

Metasploit was used to exploit the Microsoft SQL Server service running on the target machine. After configuring the exploit module and payload, a reverse Meterpreter session was obtained.

Step 2: System Enumeration

After gaining a Meterpreter shell, directory listing commands were used to explore the filesystem and identify potential flag locations.

Step 3: Retrieving Flag 1

Flag1 was found in the root directory of the C drive and accessed using the cat command.

Step 4: Retrieving Flag 2

The second flag was located inside the Windows configuration directory.

Step 5: Retrieving Flag 3

A file located in the system drivers etc directory contained the third flag and a hint regarding privilege escalation.

Step 6: Retrieving Flag 4

The final flag was located on the Administrator desktop directory.

Recovered Flags

Flag	Location	Value
Flag 1	C:\flag1.txt	2d44ee75d69b411c85e68addf2e39afe
Flag 2	C:\Windows\System32\config\flag2.txt	13deafe14d0841e4a0c6a789aba37d75
Flag 3	C:\Windows\System32\drivers\etc\EscaltePrivilageToGetThisFlag.txt	9269465288744cd6b8f07f4f83472239

Flag 4 C:\Users\Administrator\Desktop\flag4.txt eb7fe53e473f40f69fc92d80b566b40d

Screenshot Evidence:

```
[-] stdapi_fs_chdir: Operation failed: The system cannot find the file specified.
meterpreter > cd C:\Windows\System32\drivers\etc
[-] stdapi_fs_chdir: Operation failed: The system cannot find the file specified.
meterpreter > pwd
C:\
meterpreter > cd windows
meterpreter > cd system32
meterpreter > cd drivers
meterpreter > cd etc
meterpreter > ls
Listing: C:\windows\system32\drivers\etc
========================================

Mode              Size   Type  Last modified              Name
----              ----   ----  -------------              ----
100666/rw-rw-rw-  34     fil   2026-03-14 15:38:31 +0530  EscaltePrivilageToGetThisFlag.txt
100666/rw-rw-rw-  824    fil   2013-08-22 18:55:41 +0530  hosts
100666/rw-rw-rw-  3683   fil   2013-08-22 21:08:13 +0530  lmhosts.sam
100666/rw-rw-rw-  407    fil   2013-08-22 18:55:41 +0530  networks
100666/rw-rw-rw-  1358   fil   2013-08-22 18:55:41 +0530  protocol
100666/rw-rw-rw-  17463  fil   2013-08-22 18:55:41 +0530  services

meterpreter > cat EscaltePrivilageToGetThisFlag.txt
9269465288744cd6b8f07f4f83472239
meterpreter >
```

Screenshot Evidence:

```
100666/rw-rw-rw-  8192       fil   2013-08-22 18:55:30 +0530  SECURITY.LOG2
100666/rw-rw-rw-  89653248   fil   2025-01-09 12:43:03 +0530  SOFTWARE
100666/rw-rw-rw-  0          fil   2013-08-22 18:55:30 +0530  SOFTWARE.LOG
100666/rw-rw-rw-  12288      fil   2013-08-22 18:55:30 +0530  SOFTWARE.LOG1
100666/rw-rw-rw-  335872     fil   2013-08-22 18:55:30 +0530  SOFTWARE.LOG2
100666/rw-rw-rw-  12582912   fil   2025-01-09 12:43:03 +0530  SYSTEM
100666/rw-rw-rw-  0          fil   2013-08-22 18:55:30 +0530  SYSTEM.LOG
100666/rw-rw-rw-  12460032   fil   2013-08-22 18:55:30 +0530  SYSTEM.LOG1
100666/rw-rw-rw-  4153344    fil   2013-08-22 18:55:30 +0530  SYSTEM.LOG2
100666/rw-rw-rw-  65536      fil   2021-09-15 11:22:34 +0530  SYSTEM{9c9f9f4c-ae7e-11e3-80ba-0026b955d
100666/rw-rw-rw-  524288     fil   2021-09-15 11:22:34 +0530  SYSTEM{9c9f9f4c-ae7e-11e3-80ba-0026b955d
100666/rw-rw-rw-  524288     fil   2021-09-15 11:22:34 +0530  SYSTEM{9c9f9f4c-ae7e-11e3-80ba-0026b955d
040777/rwxrwxrwx  8192       dir   2014-03-18 16:02:48 +0530  TxR
100666/rw-rw-rw-  34         fil   2026-03-14 15:38:31 +0530  flag2.txt
040777/rwxrwxrwx  4096       dir   2014-06-21 01:26:20 +0530  systemprofile

meterpreter > cat flag2.txt
13deafe14d0841e4a0c6a789aba37d75
meterpreter >
```

Screenshot Evidence:

```
meterpreter > cd Desktop
meterpreter > ls
Listing: C:\users\Administrator\Desktop
=======================================

Mode           Size  Type  Last modified           Name
----           ----  ----  -------------           ----
100666/rw-rw-rw-  282   fil   2025-01-09 12:42:42 +0530  desktop.ini
100666/rw-rw-rw-  34    fil   2026-03-14 15:38:31 +0530  flag4.txt

meterpreter > cat flag4.txt
eb7fe53e473f40f69fc92d80b566b40d
meterpreter >
```

Screenshot Evidence:

```
meterpreter > cd /
meterpreter > ls
Listing: C:\
============

Mode              Size    Type  Last modified              Name
----              ----    ----  -------------              ----
040777/rwxrwxrwx  0       dir   2021-12-15 09:58:20 +0530  $Recycle.Bin
100666/rw-rw-rw-  1       fil   2013-06-18 17:48:29 +0530  BOOTNXT
040777/rwxrwxrwx  0       dir   2013-08-22 20:18:41 +0530  Documents and Settings
040777/rwxrwxrwx  0       dir   2013-08-22 21:22:33 +0530  PerfLogs
040555/r-xr-xr-x  4096    dir   2025-01-09 12:30:38 +0530  Program Files
040777/rwxrwxrwx  4096    dir   2024-12-15 14:57:59 +0530  Program Files (x86)
040777/rwxrwxrwx  4096    dir   2015-08-13 21:42:59 +0530  ProgramData
040777/rwxrwxrwx  0       dir   2021-12-31 13:30:32 +0530  System Volume Information
040555/r-xr-xr-x  4096    dir   2025-01-09 12:42:28 +0530  Users
040777/rwxrwxrwx  24576   dir   2025-01-09 12:38:38 +0530  Windows
100444/r--r--r--  398356  fil   2014-03-18 15:35:18 +0530  bootmgr
100666/rw-rw-rw-  34      fil   2026-03-14 15:38:31 +0530  flag1.txt
000000/----------  0       fif   1970-01-01 05:30:00 +0530  pagefile.sys

meterpreter > cat flag1.txt
2d44ee75d69b411c85e68addf2e39afe
meterpreter >
```